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CONTACT INFORMATION

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ACADEMIC POSITIONS

Postdoctoral Fellow, The University of Hong Kong	2025 - Present
Adam Smith Career Development Fellow, University of Glasgow	2023-2025

EDUCATION

Syracuse University	2023
Ph.D. in Economics	
Duke University	2018
M.A. in Economics	
Nankai University	2015
Bachelor of Economics	

RESEARCH INTERESTS

Applied Microeconomics (Labor, Environmental, and Urban); Applied Econometrics

PUBLICATIONS

“Heat and Productivity: Evidence from Flight On-Time Performance” with Ying Shi
Journal of the Association of Environmental and Resource Economists 12, no. 5 (2025): 1243-1275. <https://doi.org/10.1086/733369>

“Monopsony in Academia and the Gender Pay Gap: Evidence from California” with Alfonso Flores-Lagunes
Labour Economics 100, June 2026, 102885. <https://doi.org/10.1016/j.labeco.2026.102885>

Human Capital and Labor Market Effects of Elite Schooling in the United Kingdom: New Causal Evidence from Nonparametric Bounds with Vikesh Amin, Carlos A. Flores, Alfonso Flores-Lagunes, and Giuseppe Germinario
Forthcoming in Research in Labor Economics, Human Capital and Economic Outcomes: Evidence, Institutions and Policy.

WORKING PAPER

“Rerouting Externalities: Flight Diversions and Local Health” with Alex Yu

Transportation networks absorb disruptions by rerouting activity, shifting local externalities toward communities that did not generate the disruption. We study this rerouting mechanism through flight diversions, linking flight diversion records to neighborhood Emergency Medical Services dispatches in New York City. Using disruptions elsewhere in the air network as an instrument for diversion exposure, we find that greater exposure increases EMS demand, especially in neighborhoods closest to airports and late-night diversions. Delayed scheduled arrivals generate markedly different responses, indicating that spatial reallocation, rather than late arrival alone, contributes importantly to the EMS response. The 2016 relaxation of slot controls at Newark Airport was followed by diversion activity shifting toward the region’s other major airports, and higher EMS demand in nearby communities. The findings illustrate how disruptions and airport-capacity policies can redistribute aircraft activity and associated health burdens across connected airports and surrounding communities.

“Carbon Footprint of Place-Based Economic Policies” with Yao Wang and Sayahnika Basu

We evaluate the unintended environmental impacts of Special Economic Zones (SEZs), a place-based policy aimed at promoting economic development in India and characterized by non-energy related tax holidays, on firms’ energy use and carbon emissions. Using detailed firm level energy data and a Difference-in-Differences design combined with matching, we find that firms located within SEZs reduce their carbon emissions by 22% compared to comparable firms outside SEZs. This reduction is driven primarily by the adoption of newer, cleaner capital rather than by declines in output. We also find suggestive evidence of a shift in energy composition away from conventional fossil fuels toward cleaner energy sources. Emission reductions are larger among bigger firms, non-manufacturing firms with greater energy-substitution flexibility, and firms in regions with better access to clean energy infrastructure.

“Extreme Temperature and Labor Market Reallocation: Spatial Mobility and Employment Adjustment” with Jindong Pang

Climate change may reshape labor markets not only by reducing productivity but also by inducing workers to adjust where and how they work. This paper examines whether and how households adapt to extreme temperature shocks through spatial mobility and employment reallocation. Using longitudinal data from China, we study the effects of extreme temperatures on internal migration and transitions between formal and informal employment. We find that extreme heat significantly increases inter-provincial migration: a one-standard-deviation increase in extreme high-temperature exposure raises the probability of moving across provinces by 0.9 percentage points, or 32% relative to the mean. In contrast, we find no detectable effect on intra-provincial migration and substantially smaller effects of extreme cold. Beyond geographic mobility, extreme heat also induces employment reallocation: a one-standard-deviation increase in heat exposure reduces the probability of transitioning from informal to formal employment by 4 percentage points (36%) and increases the probability of transitioning from formal to informal employment by 3 percentage points (23%). Heterogeneous responses across demographic and industry groups suggest that these patterns cannot be explained by productivity losses alone and reflect broader adjustments in labor-market choices. Our findings identify geographic mobility and employment transitions as important but uneven margins of climate adaptation, highlighting the unequal capacity of workers to respond to rising temperatures.

“The Changing Monopsony Power in Higher Education: Evidence without Instruments” with Linqi Zhang

This study investigates the dynamics of universities’ wage-setting power in a public system located in a “red” state that has undergone major tenure-policy changes. Facing challenges due to the absence of valid instruments and with an endogenous, potentially mismeasured salary variable, we propose a method to estimate monopsony power without instruments while accounting for measurement error in the endogenous regressor. We find substantial wage-setting power, exceeding both the national average and that of a comparable public system in a “blue” state. The estimated monopsony power declined during the policy-change period and rebounded in the COVID period.

“Air Pollution, Workplace Safety, and Productivity: Evidence Beyond LATE”

This paper studies the causal effect of air pollution on workplace safety in the United States using novel nationwide administrative data on severe workplace injuries. Moving beyond the standard method that exploits instrumental variables to address endogenous air pollution and identify the local average treatment effect (LATE) for compliers, this study applies partial identification approaches based on weaker assumptions to recover informative bounds on the average treatment effect (ATE). I find that PM2.5 pollution raises the incidence of severe workplace

accidents. Counterfactual reductions in high-pollution days yield sizable economic benefits, equivalent to at least 9% of the EPA's annual air-pollution control expenditures.

SELECTED WORK IN PROGRESS

“The Effect of Tax Levies on Future Construction and Demolitions: The Importance of Zeros When Leveraging Voting Designs” with *David Brasington* and *Alfonso Flores-Lagunes*

“Gender Gaps in Research Productivity Metrics” with *Alfonso Flores-Lagunes*

SEMINAR & CONFERENCE

2026 The 5th Asian Economic Development Conference; The 4th Summer Meeting in Urban Economics, China

2025 The Allied Social Science Associations (ASSA) Annual meeting (presented by Alfonso Flores-Lagunes); Duke Kunshan University; The Annual Conference of the International Association for Applied Econometrics; The 3rd Summer Meeting in Urban Economics, China

2024 The Econometric Society European Winter Meeting; The Forum for Research on Eastern Europe: Climate and Environment Workshop, Stockholm School of Economics; UC San Diego 21st Century India Center Virtual Seminar Series; European Association of Environmental and Resource Economists Annual Conference; Scottish Economic Society Annual Conference; Royal Economic Society Annual Conference; Eastern Economic Association Annual Conference (AERE-sponsored session)

2023 Canadian Economics Association Conference; Western Economic Association Annual Conference (AERE-sponsored session); Association of Environmental and Resource Economists Summer Conference; American Society of Health Economists Annual Conference; The Lewin Group; The Ohio State University; University of Glasgow

2022 Southern Economics Association Annual Conference; Syracuse University Applied Micro Seminar; Midwest Economics Association the 86th Annual Meetings (Joint Session with the Society of Labor Economists)

REFEREING SERVICE

Journal of Environmental Economics and Management (1), Journal of Health Economics (1), International Journal of Industrial Organization (1), Regional Science and Urban Economics (1), Journal of Population Economics (4), Review of Development Economics (1), Environmental and Resource Economics (1), Environmental Science and Pollution Research (1), Education Economics (1), China Economic Review (2)

TEACHING EXPERIENCE

Undergraduate Level

Primary Instructor

Introduction to Statistics and Econometrics (Online), Syracuse University 2021
Course Feedback: 5.67/6

Teaching Assistant

Economic Statistics, Syracuse University 2021
Labor Economics, Syracuse University 2021
Introduction to Statistics and Econometrics, Syracuse University 2020
Intermediate Microeconomics, Syracuse University 2018; 2020
Introductory Microeconomics, Syracuse University 2019
Economic Ideas and Issues, Syracuse University 2019

Dissertation Supervisor

6 students, Economics Subject, University of Glasgow AY 2023-2024

Graduate Level

Co-Instructor

Microeconometrics: Impact Evaluation and Causal Analysis,
University of Glasgow Fall 2023; Fall 2024

Teaching Assistant

Mathematics for Economists, Syracuse University Summer 2019

	Introduction to Mathematical Statistics, Duke University	Fall 2017
RESEARCH EXPERIENCE	<i>Research Assistant</i> Alfonso Flores-Lagunes; Maria Zhu Syracuse University Jisung Park University of California, Los Angeles Sharon Belenzon; John Graham Duke University	2021-2022 2018 2017-2018
AWARDS & SCHOLARSHIPS	Research Excellence Doctoral Funding, Syracuse University Maxwell School Summer Fellowship, Syracuse University Graduate Assistantship, Syracuse University The M.A. Merit Scholar Award, Duke University Hezhan Scholarship, Nankai University The Excellent Undergraduate Scholarship, Nankai University	2021-2022 2018-2022 2018-2020 2017 2012; 2014 2013
SKILLS & LANGUAGES	Programming: Stata, Python, R, $\LaTeX$, ArcGIS, MATLAB Languages: Mandarin (Native), English (Fluent)	

Last Updated: October 7, 2026